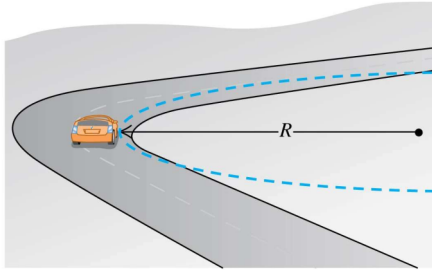


Example (a). Flat Curved Road



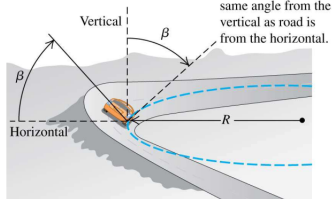
Need a centripetal force to stay 'on track' (i.e. move along an arc of a circle)". This is provided by static friction between tyres and the road.

The maximum speed at which the car does not skid can be found by taking the maximum magnitude of static friction

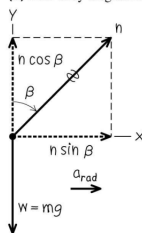
$$\frac{mv_{\max}^2}{R} = \mu_s mg \quad \Rightarrow \quad v_{\max} = \sqrt{\frac{\mu_s g R}{R}}.$$

Example (b). Banked Curved Road. No Friction

(a) Car rounding banked curve



(b) Free-body diagram for car



At a given speed v and radius of curvature R , there is an angle for which static friction is not necessary and the normal force is sufficient to provide the necessary centripetal force. This angle is determined by the conditions¹

$$\frac{mv^2}{R} = n \sin \beta \quad \text{and} \quad w = n \cos \beta.$$

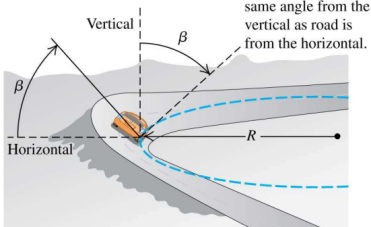
Dividing these equations side by side, we get

$$\frac{v^2}{Rg} = \tan \beta.$$

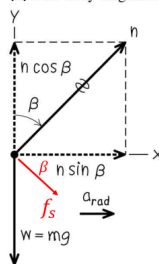
¹Caution! Here $n \neq w \cos \beta$, unlike in the examples with inclined planes discussed before. It is a different situation here, as the car is moving along a curve and not simply sliding on an inclined plane. Therefore, here we have $n > w$.

Example (c). Banked Curved Road. Friction

(a) Car rounding banked curve



(b) Free-body diagram for car



Friction can assist the centripetal force and allow for a higher speed when turning on a banked surface. The car does not skid if

$$\frac{mv^2}{R} = n \sin \beta + f_s \cos \beta \quad \text{and} \quad w = n \cos \beta - f_s \sin \beta.$$

For the maximum possible turning speed, the magnitude of static friction is maximum $f_s = \mu_s n$. Hence

$$\frac{mv^2}{R} = n \sin \beta + \mu_s n \cos \beta \quad \text{and} \quad w = n \cos \beta - \mu_s n \sin \beta.$$

Example (c). Banked Curved Road. Friction

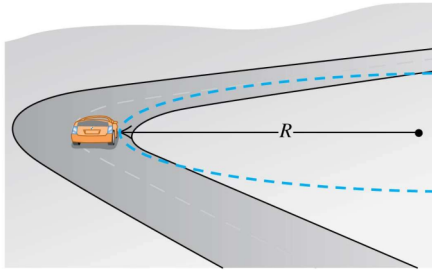
Dividing both equations side by side

$$\frac{v_{\max}^2}{Rg} = \frac{n \sin \beta + \mu_s n \cos \beta}{n \cos \beta - \mu_s n \sin \beta} = \frac{\sin \beta + \mu_s \cos \beta}{\cos \beta - \mu_s \sin \beta}.$$

Eventually, the maximum speed at which the car can take the curve is

$$v_{\max} = \sqrt{\frac{\tan \beta + \mu_s}{1 - \mu_s \tan \beta} Rg}.$$

Example (a). Flat Curved Road

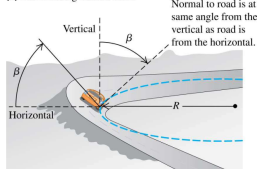


Inertial observer: "Need a centripetal force to stay 'on track' (i.e. move along an arc of a circle). This is provided by static friction between tyres and the road."

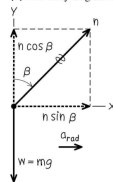
Non-inertial observer (at the center of the circle; rotating along with the moving car): "The centrifugal 'force' is balanced by static friction between tyres and the road."

Examples (b) and (c). Banked Curved Road

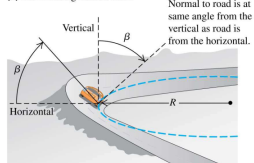
(a) Car rounding banked curve



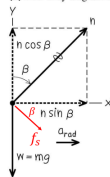
(b) Free-body diagram for car



(a) Car rounding banked curve



(b) Free-body diagram for car



Exercise: Rephrase the discussion in words of a non-inertial observer (located at the center of the circle; rotating along with the moving car).